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# PWM Fan Speed Controller Using a 555 Timer

By: Ian Hall

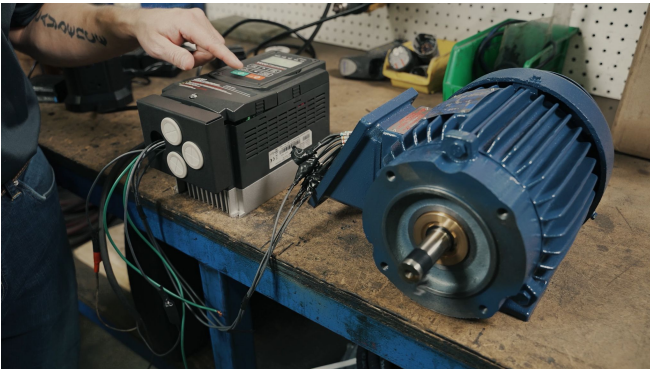
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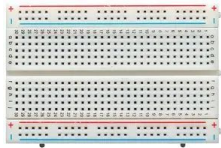
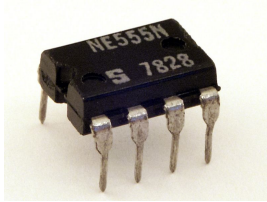
# Motivation



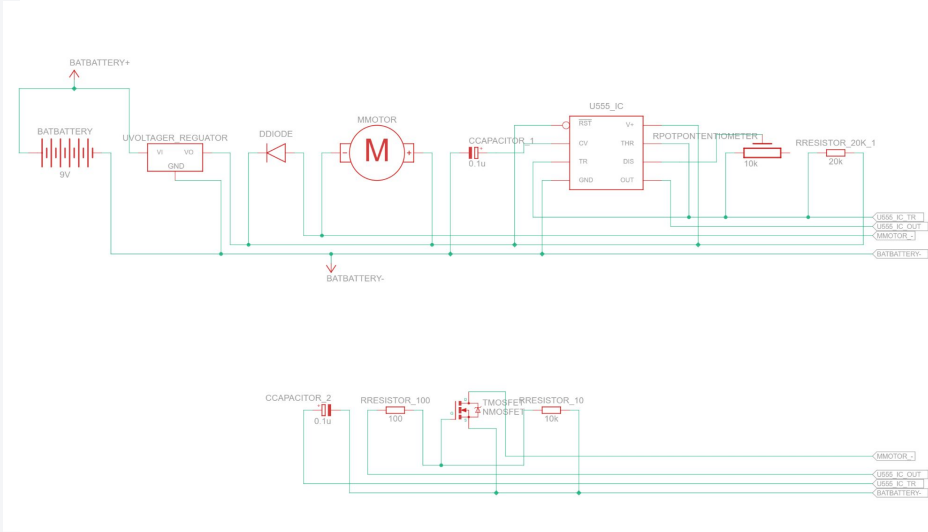
- Need to control fan speed efficiently
- Traditional methods waste energy as heat
- PWM provides efficient power control
- Used in real systems (PC cooling, motors)
- Goal: build a simple, adjustable speed controller



# Parts

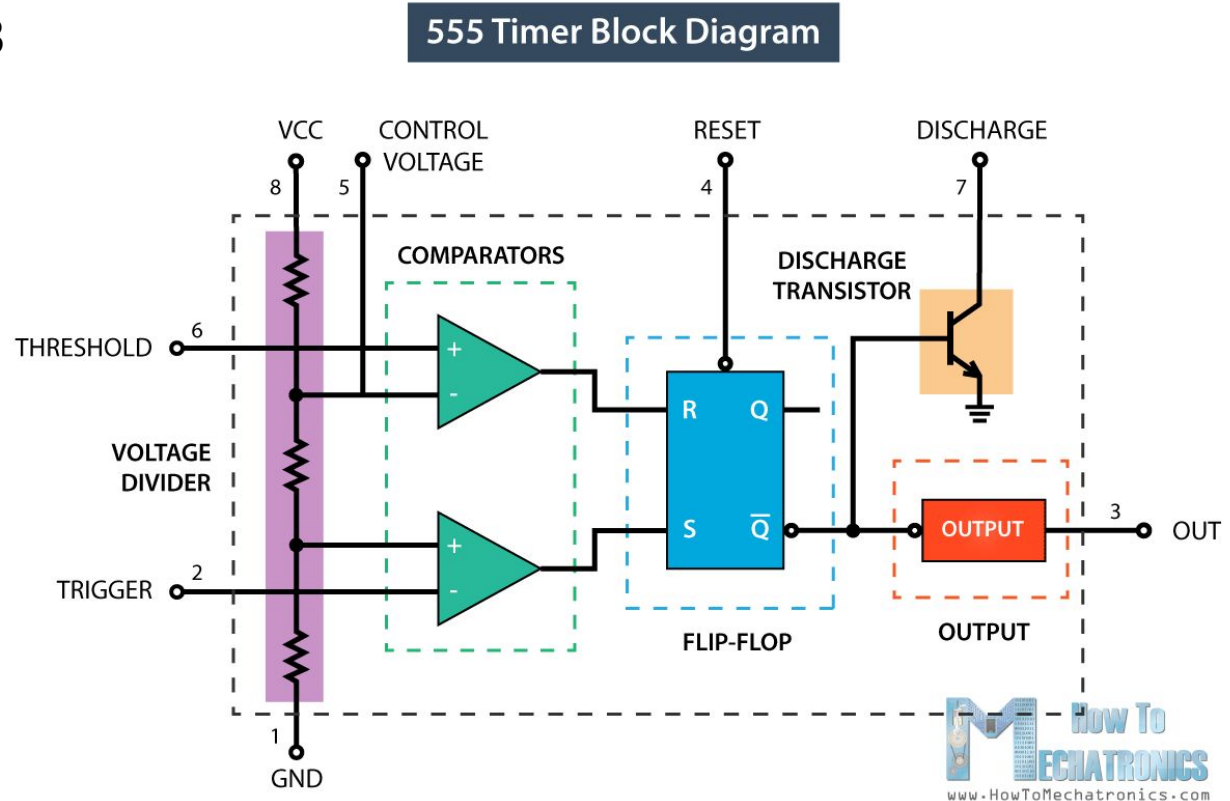


- 555 Timer IC
- Resistors(20k, 100, 10k)
- Potentiometer(10k)
- Capacitor(0.1uF)
- MOSFET IRLZ44N
- Diode
- Fan (5v)
- 5V Power supply
- Breadboard
- Jumper wires



# How the 555 IC Timer works

- Internal divider sets  $1/3$   $V_{cc}$  and  $2/3$   $V_{cc}$
- Pins 2 & 6 monitor capacitor voltage
- Comparators drive the flip-flop
- Flip-flop controls Pin 3 and Pin 7
- Pin 4 keeps the timer enabled
- Pin 5 improves noise stability
- Cycle repeats to create PWM



# PWM & Duty Cycle

## Pulse Width Modulation

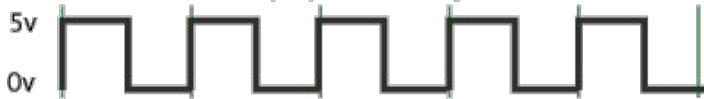
0% Duty Cycle - `analogWrite(0)`



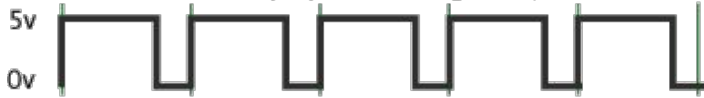
25% Duty Cycle - `analogWrite(64)`



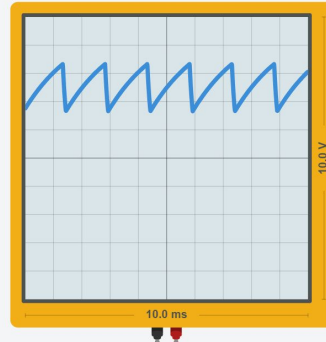
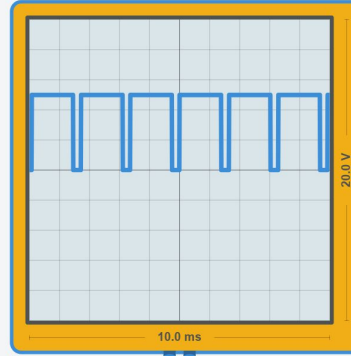
50% Duty Cycle - `analogWrite(127)`



75% Duty Cycle - `analogWrite(191)`

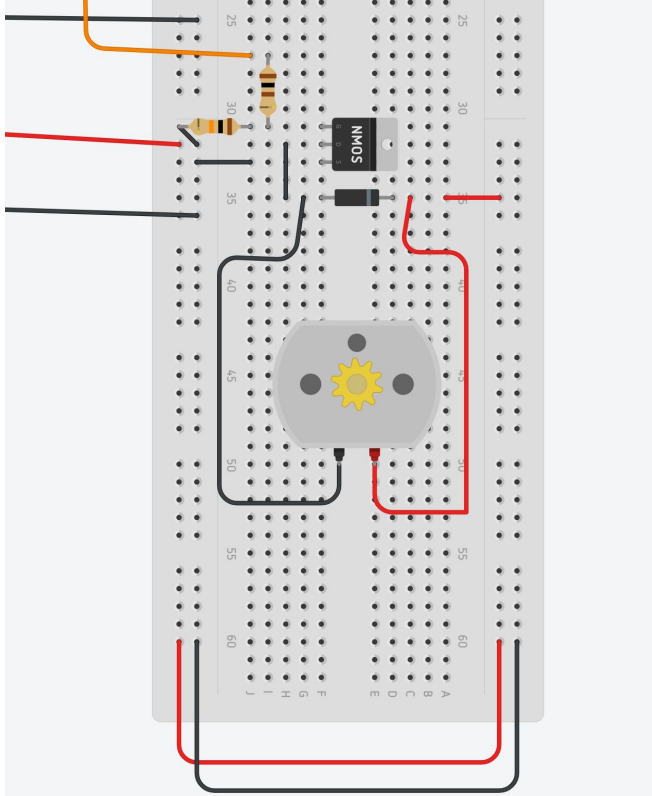


100% Duty Cycle - `analogWrite(255)`



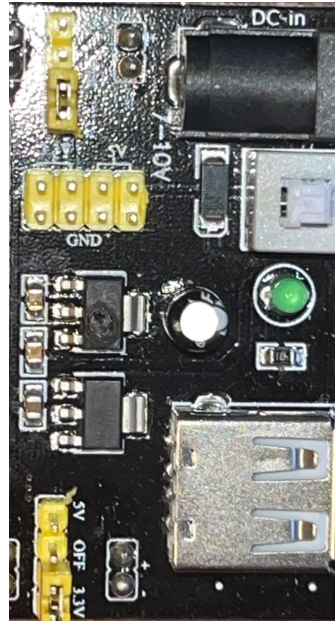
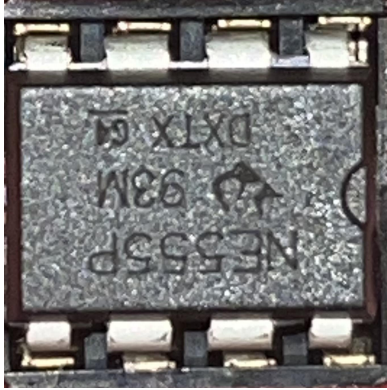
- PWM switches voltage ON and OFF rapidly
- Duty cycle = % of time signal is ON
- Higher duty cycle → higher average voltage/power
- Lower duty cycle → lower average voltage/power
- Capacitor charges and discharges to create timing
- In this circuit, duty cycle controls fan speed

# MOSFET + FAN + DIODE



- MOSFET acts as an electronic switch
- Gate voltage from 555 controls ON/OFF switching
- When ON: current flows through fan to ground
- When OFF: current flow stops
- Diode protects circuit from voltage spikes (back EMF)
- Fan speed depends on PWM switching from MOSFET

# Challenges



- Applied excessive voltage → damaged power module and IC
- MOSFET gate floating → fan stuck ON
- Loose 555 IC → loss of PWM control signal
- Limited speed range → resistor values not optimal
- Breadboard wiring



# Fixes & Questions

- Why use PWM?
  - Efficient power control (less heat loss)
- Why use a MOSFET?
  - Allows low-power signal to control high current
- Why use a diode?
  - Protects circuit from inductive voltage spikes
- Replaced damaged power module and components
- Added pull-down resistor to stabilize MOSFET gate
- Secured IC and connections
- Increased resistor value (10k → 20k) for better control range

# IRL circuit + video

